

README

Lab 5 - Food Delivery Order Status Classification

Project overview

This notebook explores a food delivery dataset and builds a Random Forest classifier to predict order status. It prepares the data, evaluates a baseline model, and compares feature engineering and feature selection experiments.

What the notebook covers

Task 1 - Delivery efficiency	Creates <code>order_efficiency</code> from actual and estimated delivery time.
Task 2 - Peak hours	Changes the peak-hour definition and compares model accuracy with the original result.
Task 3 - Top-K features	Uses Random Forest feature importance to compare models trained with the top 10, 20, 30, and 50 features.
Task 4 - Feature selection	Uses <code>SelectFromModel</code> with Random Forest and compares accuracy before and after selection.

How to use

Open **lab5_completed.ipynb** in Jupyter Notebook, JupyterLab, or Google Colab. Run the cells from top to bottom so that the dataset, preprocessing steps, and model variables are defined before the experiments that use them.

Notebook contents

The notebook includes data preparation, train/test splitting, a baseline Random Forest model, feature importance, `SelectFromModel` selection, Top-K comparisons, and a final accuracy comparison. The explanatory Task 2, Task 3, and Task 4 text cells have been added to the relevant sections.

Files

Notebook	lab5_completed.ipynb
Guide	README_lab5.pdf